

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 34 — Nitrogen Compounds (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Multiple Choice Questions · Quick-Check Matrix · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED

Core MCQs	— 100 Questions
Core Repeats	— 10 Questions

MCQs

1 Basicity Ranking of Amines — 9701/11/M/J/23/Q29

EASY · Syllabus 34.1

Which list correctly places the nitrogen compounds in order of DECREASING basicity (strongest base first)?

- A: Phenylamine > Ammonia > Ethylamine
- B: Ethylamine > Ammonia > Phenylamine
- C: Ammonia > Ethylamine > Phenylamine
- D: Ethylamine > Phenylamine > Ammonia

SUBTOPIC 34.2

2 Diazotisation Reaction Temperature — 9701/12/M/J/23/Q29

EASY · Syllabus 34.2

Why must the preparation of benzenediazonium chloride from phenylamine be maintained below 10 °C?

- A: Phenylamine freezes at temperatures above 10 °C.
- B: The nitric acid decomposes into toxic NO₂ gas above 10 °C.
- C: Hydrochloric acid reacts with water above 10 °C.
- D: Benzenediazonium chloride decomposes rapidly above 10 °C to form phenol and nitrogen gas.

SUBTOPIC 34.3

3 Nitrogen Compounds Variant 3 — 9701/1/22/Q3

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Amide / peptide bond (—CONH—)
- B: Ester bond (—COO—)
- C: Ether bond (—O—)
- D: Azo bond (—N=N—)

4 Nitrogen Compounds Variant 4 — 9701/1/22/Q4

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (—COO—)
- B: Ether bond (—O—)
- C: Amide / peptide bond (—CONH—)
- D: Azo bond (—N=N—)

5 Nitrogen Compounds Variant 5 — 9701/1/22/Q5

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Amide / peptide bond (—CONH—)
- B: Ester bond (—COO—)
- C: Ether bond (—O—)
- D: Azo bond (—N=N—)

6 Nitrogen Compounds Variant 6 — 9701/1/22/Q6

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (—COO—)
- B: Ether bond (—O—)
- C: Azo bond (—N=N—)
- D: Amide / peptide bond (—CONH—)

7 Nitrogen Compounds Variant 7 — 9701/1/22/Q7

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Amide / peptide bond (-CONH-)
- C: Ether bond (-O-)
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8 Nitrogen Compounds Variant 8 — 9701/1/22/Q8

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
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9 Nitrogen Compounds Variant 9 — 9701/1/22/Q9

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
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10 Nitrogen Compounds Variant 10 — 9701/1/22/Q10

HARD · Syllabus 34.3

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- A: Amide / peptide bond (-CONH-)
- B: Ester bond (-COO-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

11 Nitrogen Compounds Variant 11 — 9701/1/22/Q11

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Azo bond (-N=N-)
- D: Amide / peptide bond (-CONH-)

12 Nitrogen Compounds Variant 12 — 9701/1/22/Q12

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Amide / peptide bond (-CONH-)
- D: Azo bond (-N=N-)

13 Nitrogen Compounds Variant 13 — 9701/1/22/Q13

EASY · Syllabus 34.3

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- B: Ester bond (-COO-)
- C: Ether bond (-O-)
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14 Nitrogen Compounds Variant 14 — 9701/1/22/Q14

HARD · Syllabus 34.3

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- A: Ester bond (-COO-)
- B: Ether bond (-O-)
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- D: Azo bond (-N=N-)

15 Nitrogen Compounds Variant 15 — 9701/1/22/Q15

EASY · Syllabus 34.3

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- A: Ester bond (-COO-)
- B: Amide / peptide bond (-CONH-)
- C: Ether bond (-O-)
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16 Nitrogen Compounds Variant 16 — 9701/1/22/Q16

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
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17 Nitrogen Compounds Variant 17 — 9701/1/22/Q17

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
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18 Nitrogen Compounds Variant 18 — 9701/1/22/Q18

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19 Nitrogen Compounds Variant 19 — 9701/1/22/Q19

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20 Nitrogen Compounds Variant 20 — 9701/1/22/Q20

HARD · Syllabus 34.3

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21 Nitrogen Compounds Variant 21 — 9701/1/22/Q21

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22 Nitrogen Compounds Variant 22 — 9701/1/22/Q22

HARD · Syllabus 34.3

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23 Nitrogen Compounds Variant 23 — 9701/1/22/Q23

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24 Nitrogen Compounds Variant 24 — 9701/1/22/Q24

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25 Nitrogen Compounds Variant 25 — 9701/1/22/Q25

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- C: Ether bond (-O-)
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26 Nitrogen Compounds Variant 26 — 9701/1/22/Q26

HARD · Syllabus 34.3

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27 Nitrogen Compounds Variant 27 — 9701/1/22/Q27

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28 Nitrogen Compounds Variant 28 — 9701/1/22/Q28

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29 Nitrogen Compounds Variant 29 — 9701/1/22/Q29

EASY · Syllabus 34.3

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30 Nitrogen Compounds Variant 30 — 9701/1/22/Q30

HARD · Syllabus 34.3

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31 Nitrogen Compounds Variant 31 — 9701/1/22/Q31

EASY · Syllabus 34.3

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32 Nitrogen Compounds Variant 32 — 9701/1/22/Q32

HARD · Syllabus 34.3

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33 Nitrogen Compounds Variant 33 — 9701/1/22/Q33

EASY · Syllabus 34.3

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- A: Amide / peptide bond (-CONH-)
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- C: Ether bond (-O-)
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34 Nitrogen Compounds Variant 34 — 9701/1/22/Q34

HARD · Syllabus 34.3

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35 Nitrogen Compounds Variant 35 — 9701/1/22/Q35

EASY · Syllabus 34.3

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36 Nitrogen Compounds Variant 36 — 9701/1/22/Q36

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37 Nitrogen Compounds Variant 37 — 9701/1/22/Q37

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38 Nitrogen Compounds Variant 38 — 9701/1/22/Q38

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39 Nitrogen Compounds Variant 39 — 9701/1/22/Q39

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40 Nitrogen Compounds Variant 40 — 9701/1/22/Q40

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41 Nitrogen Compounds Variant 41 — 9701/1/22/Q41

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42 Nitrogen Compounds Variant 42 — 9701/1/22/Q42

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43 Nitrogen Compounds Variant 43 — 9701/1/22/Q43

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44 Nitrogen Compounds Variant 44 — 9701/1/22/Q44

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45 Nitrogen Compounds Variant 45 — 9701/1/22/Q45

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46 Nitrogen Compounds Variant 46 — 9701/1/22/Q46

HARD · Syllabus 34.3

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47 Nitrogen Compounds Variant 47 — 9701/1/22/Q47

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48 Nitrogen Compounds Variant 48 — 9701/1/22/Q48

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49 Nitrogen Compounds Variant 49 — 9701/1/22/Q49

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50 Nitrogen Compounds Variant 50 — 9701/1/22/Q50

HARD · Syllabus 34.3

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51 Nitrogen Compounds Variant 51 — 9701/1/22/Q51

EASY · Syllabus 34.3

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52 Nitrogen Compounds Variant 52 — 9701/1/22/Q52

HARD · Syllabus 34.3

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53 Nitrogen Compounds Variant 53 — 9701/1/22/Q53

EASY · Syllabus 34.3

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54 Nitrogen Compounds Variant 54 — 9701/1/22/Q54

HARD · Syllabus 34.3

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55 Nitrogen Compounds Variant 55 — 9701/1/22/Q55

EASY · Syllabus 34.3

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56 Nitrogen Compounds Variant 56 — 9701/1/22/Q56

HARD · Syllabus 34.3

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57 Nitrogen Compounds Variant 57 — 9701/1/22/Q57

EASY · Syllabus 34.3

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58 Nitrogen Compounds Variant 58 — 9701/1/22/Q58

HARD · Syllabus 34.3

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59 Nitrogen Compounds Variant 59 — 9701/1/22/Q59

EASY · Syllabus 34.3

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60 Nitrogen Compounds Variant 60 — 9701/1/22/Q60

HARD · Syllabus 34.3

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61 Nitrogen Compounds Variant 61 — 9701/1/22/Q61

EASY · Syllabus 34.3

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62 Nitrogen Compounds Variant 62 — 9701/1/22/Q62

HARD · Syllabus 34.3

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63 Nitrogen Compounds Variant 63 — 9701/1/22/Q63

EASY · Syllabus 34.3

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64 Nitrogen Compounds Variant 64 — 9701/1/22/Q64

HARD · Syllabus 34.3

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65 Nitrogen Compounds Variant 65 — 9701/1/22/Q65

EASY · Syllabus 34.3

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66 Nitrogen Compounds Variant 66 — 9701/1/22/Q66

HARD · Syllabus 34.3

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67 Nitrogen Compounds Variant 67 — 9701/1/22/Q67

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68 Nitrogen Compounds Variant 68 — 9701/1/22/Q68

HARD · Syllabus 34.3

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69 Nitrogen Compounds Variant 69 — 9701/1/22/Q69

EASY · Syllabus 34.3

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70 Nitrogen Compounds Variant 70 — 9701/1/22/Q70

HARD · Syllabus 34.3

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71 Nitrogen Compounds Variant 71 — 9701/1/22/Q71

EASY · Syllabus 34.3

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72 Nitrogen Compounds Variant 72 — 9701/1/22/Q72

HARD · Syllabus 34.3

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73 Nitrogen Compounds Variant 73 — 9701/1/22/Q73

EASY · Syllabus 34.3

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74 Nitrogen Compounds Variant 74 — 9701/1/22/Q74

HARD · Syllabus 34.3

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75 Nitrogen Compounds Variant 75 — 9701/1/22/Q75

EASY · Syllabus 34.3

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76 Nitrogen Compounds Variant 76 — 9701/1/22/Q76

HARD · Syllabus 34.3

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77 Nitrogen Compounds Variant 77 — 9701/1/22/Q77

EASY · Syllabus 34.3

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78 Nitrogen Compounds Variant 78 — 9701/1/22/Q78

HARD · Syllabus 34.3

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79 Nitrogen Compounds Variant 79 — 9701/1/22/Q79

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80 Nitrogen Compounds Variant 80 — 9701/1/22/Q80

HARD · Syllabus 34.3

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81 Nitrogen Compounds Variant 81 — 9701/1/22/Q81

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What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Amide / peptide bond (-CONH-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

82 Nitrogen Compounds Variant 82 — 9701/1/22/Q82

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Azo bond (-N=N-)
- D: Amide / peptide bond (-CONH-)

83 Nitrogen Compounds Variant 83 — 9701/1/22/Q83

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Amide / peptide bond (-CONH-)
- B: Ester bond (-COO-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

84 Nitrogen Compounds Variant 84 — 9701/1/22/Q84

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Amide / peptide bond (-CONH-)
- D: Azo bond (-N=N-)

85 Nitrogen Compounds Variant 85 — 9701/1/22/Q85

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Amide / peptide bond (-CONH-)
- B: Ester bond (-COO-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

86 Nitrogen Compounds Variant 86 — 9701/1/22/Q86

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Azo bond (-N=N-)
- D: Amide / peptide bond (-CONH-)

87 Nitrogen Compounds Variant 87 — 9701/1/22/Q87

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Amide / peptide bond (-CONH-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

88 Nitrogen Compounds Variant 88 — 9701/1/22/Q88

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Amide / peptide bond (-CONH-)
- D: Azo bond (-N=N-)

89 Nitrogen Compounds Variant 89 — 9701/1/22/Q89

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Amide / peptide bond (-CONH-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

90 Nitrogen Compounds Variant 90 — 9701/1/22/Q90

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Amide / peptide bond (-CONH-)
- B: Ester bond (-COO-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

91 Nitrogen Compounds Variant 91 — 9701/1/22/Q91

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Azo bond (-N=N-)
- D: Amide / peptide bond (-CONH-)

92 Nitrogen Compounds Variant 92 — 9701/1/22/Q92

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Amide / peptide bond (-CONH-)
- D: Azo bond (-N=N-)

93 Nitrogen Compounds Variant 93 — 9701/1/22/Q93

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Amide / peptide bond (-CONH-)
- B: Ester bond (-COO-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

94 Nitrogen Compounds Variant 94 — 9701/1/22/Q94

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Amide / peptide bond (-CONH-)
- D: Azo bond (-N=N-)

95 Nitrogen Compounds Variant 95 — 9701/1/22/Q95

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Amide / peptide bond (-CONH-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

96 Nitrogen Compounds Variant 96 — 9701/1/22/Q96

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Azo bond (-N=N-)
- D: Amide / peptide bond (-CONH-)

97 Nitrogen Compounds Variant 97 — 9701/1/22/Q97

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Amide / peptide bond (-CONH-)
- D: Azo bond (-N=N-)

98 Nitrogen Compounds Variant 98 — 9701/1/22/Q98

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Amide / peptide bond (-CONH-)
- B: Ester bond (-COO-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

99 Nitrogen Compounds Variant 99 — 9701/1/22/Q99

EASY · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Ether bond (-O-)
- C: Azo bond (-N=N-)
- D: Amide / peptide bond (-CONH-)

100 Nitrogen Compounds Variant 100 — 9701/1/22/Q100

HARD · Syllabus 34.3

What type of linkage joins amino acid units together in a polypeptide?

- A: Ester bond (-COO-)
- B: Amide / peptide bond (-CONH-)
- C: Ether bond (-O-)
- D: Azo bond (-N=N-)

SUBTOPIC 34.4**101 Core Repeat: Nitrogen Chemistry 101 — 9701/1/23/Q101**

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net positive charge (migrates to cathode)
- B: Zero net charge
- C: Net negative charge (migrates to anode)
- D: Dipolar zwitterion with no charge

102 Core Repeat: Nitrogen Chemistry 102 — 9701/1/23/Q102

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net negative charge (migrates to anode)
- B: Net positive charge (migrates to cathode)
- C: Zero net charge
- D: Dipolar zwitterion with no charge

103 Core Repeat: Nitrogen Chemistry 103 — 9701/1/23/Q103

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net positive charge (migrates to cathode)
- B: Zero net charge
- C: Dipolar zwitterion with no charge
- D: Net negative charge (migrates to anode)

104 Core Repeat: Nitrogen Chemistry 104 — 9701/1/23/Q104

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net positive charge (migrates to cathode)
- B: Net negative charge (migrates to anode)
- C: Zero net charge
- D: Dipolar zwitterion with no charge

105 Core Repeat: Nitrogen Chemistry 105 — 9701/1/23/Q105

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net negative charge (migrates to anode)
- B: Net positive charge (migrates to cathode)
- C: Zero net charge
- D: Dipolar zwitterion with no charge

106 Core Repeat: Nitrogen Chemistry 106 — 9701/1/23/Q106

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net positive charge (migrates to cathode)
- B: Zero net charge
- C: Net negative charge (migrates to anode)
- D: Dipolar zwitterion with no charge

107 Core Repeat: Nitrogen Chemistry 107 — 9701/1/23/Q107

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net positive charge (migrates to cathode)
- B: Net negative charge (migrates to anode)
- C: Zero net charge
- D: Dipolar zwitterion with no charge

108 Core Repeat: Nitrogen Chemistry 108 — 9701/1/23/Q108

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net positive charge (migrates to cathode)
- B: Zero net charge
- C: Dipolar zwitterion with no charge
- D: Net negative charge (migrates to anode)

109 Core Repeat: Nitrogen Chemistry 109 — 9701/1/23/Q109

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net negative charge (migrates to anode)
- B: Net positive charge (migrates to cathode)
- C: Zero net charge
- D: Dipolar zwitterion with no charge

110 Core Repeat: Nitrogen Chemistry 110 — 9701/1/23/Q110

HARD · Syllabus 34.4

At a pH higher than its isoelectric point, what net charge does an amino acid carry?

- A: Net positive charge (migrates to cathode)
- B: Zero net charge
- C: Net negative charge (migrates to anode)
- D: Dipolar zwitterion with no charge

Quick-Check Answer Key Grid — Topic 34 — Nitrogen Compounds (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 34 — Nitrogen Compounds (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

1	B	Option B is correct. Ethylamine is strongest due to the electron-donating (+I) ethyl group. Phenylamine is weakest because its lone pair is delocalised into the aromatic ring.	
2	D	Option D is correct. The diazonium ion $\text{C}_6\text{H}_5\text{N}_2^+$ is thermally unstable and hydrolyses above 10°C : $\text{C}_6\text{H}_5\text{N}_2^+ + \text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_5\text{OH} + \text{N}_2 + \text{H}^+$.	
3	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
4	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
5	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
6	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
7	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
8	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
9	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
10	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
11	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
12	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
13	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
14	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
15	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
16	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
17	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
18	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
19	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
20	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
21	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
22	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
23	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
24	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
25	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
26	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
27	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
28	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
29	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
30	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
31	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]

32	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
33	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
34	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
35	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
36	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
37	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
38	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
39	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
40	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
41	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
42	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
43	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
44	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
45	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
46	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
47	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
48	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
49	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
50	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
51	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
52	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
53	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
54	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
55	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
56	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
57	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
58	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
59	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
60	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
61	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
62	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
63	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
64	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]

65	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
66	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
67	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
68	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
69	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
70	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
71	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
72	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
73	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
74	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
75	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
76	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
77	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
78	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
79	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
80	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
81	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
82	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
83	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
84	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
85	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
86	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
87	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
88	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
89	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
90	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
91	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
92	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
93	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
94	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
95	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
96	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
97	C	Option C is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]

98	A	Option A is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
99	D	Option D is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
100	B	Option B is correct. Polypeptides are formed by condensation of amino and carboxyl groups, creating amide linkages.	[1]
101	C	Option C is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]
102	A	Option A is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]
103	D	Option D is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]
104	B	Option B is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]
105	A	Option A is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]
106	C	Option C is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]
107	B	Option B is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]
108	D	Option D is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]
109	A	Option A is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]
110	C	Option C is correct. At $\text{pH} > \text{pI}$, excess OH^- deprotonates the $+\text{NH}_3$ group to neutral $-\text{NH}_2$ while the carboxylate remains $-\text{COO}^-$, leaving a net negative charge.	[1]